CaYPT

Canadian Young Physicists' Tournament

ALL DRILLING

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Abstract

In this SOP, the basic operations of drilling as well as four types of drills will be discussed. Primarily, the functions and applications of drilling.

Figure 1: Milwaukee drill and impact driver kit used as a representative cordless drilling system.

Contents

1 Principle of Operation of Drilling 4
2 Electronic Component of Drilling 4
3 Basic Characteristics and Operations of Four Types of Drilling 5
3.1 Hand Drill 5
3.2 Impact Driver 5
3.3 Drill Press 5
3.4 Tap and Die 7
4 References 8

List of Figures

Figure 1 Milwaukee drilling kit 2
Figure 2 Cordless drill battery system 4
Figure 3 Hand drill 5
Figure 4 Impact driver 5
Figure 5 Chuck key 6
Figure 6 Drill press components 6
Figure 7 Tap and die 7

1 Principle of Operation of Drilling

Drilling is a cutting process done to create a hole or circular cross-section in solid materials using a drill bit and a drill. The drill bit is usually a multi-point rotary cutting tool and, when attached to a drill, an external component spins the drill bit at high revolutions per minute. The drill then cuts through the desired area and creates the hole or circular cross-section corresponding to the size of the drill bit used.

2 Electronic Component of Drilling

The four types of drilling that will be discussed are the hand drill, impact driver, drill press, and tap and die.

Electronics: The first three of these drills are electronic drills. In our lab, the drill press is plugged in, while the hand drill and impact driver are Milwaukee branded cordless drills. They require an interchangeable battery that can be found on the bottom of the drills. There is a button on the battery pack that indicates the battery level of the battery pack in four levels when pressed. When it is on the lowest level, one bar and red, the battery can be swapped out for another by holding down the two red buttons on the battery pack and sliding it out. Then the battery pack can be slid onto the red charging kit. Compact batteries can be expected to charge in approximately 30 minutes, and extended capacity batteries in approximately 60 minutes.

Figure 2: Cordless drill kit with rechargeable batteries and charging station.

3 Basic Characteristics and Operations of Four Types of Drilling

3.1 Hand Drill

Beginning with the most standardized type of drill, the hand drill is a drill that rotates the drill bit at very high revolutions per minute when the trigger is held. To place in the desired drill bit, unscrew the inner barrel of the drill, place the drill bit, then re-screw the barrel until tight. The numbers on the drill around the barrel and on top of the main compartment are used to vary the torque. The greater the number, the greater the torque and twisting force. The button that moves from right to left beneath the main compartment changes the direction of spin on the drill. When pushed to the furthest left, the drill rotates clockwise. When pushed to the furthest right, the drill rotates counterclockwise. When the button is centered in the middle, the trigger of the drill is suppressed.

Figure 3: Hand drill.

3.2 Impact Driver

Figure 4: Impact driver.

The impact driver works the same way as the normal hand drill and has similar controls and configuration. The main difference is in the barrel component. To place a drill bit in the impact driver, pull on the right shape on its head, insert the bit, let it go, and it should stay in place. Rather than having a barrel with multiple variations in torque, the impact driver has a clean head and quick-release shanks that accept all one-quarter-inch hex driver bits. Impact drivers produce significant rotational force and are capable of driving large wood screws quickly. They produce more power and rotational force than a hand drill.

3.3 Drill Press

A drill press is essentially a drill that is oriented toward the floor and mounted perpendicular to a small surface that can support whatever surface needs to be drilled. These machines can have variable speeds, and some have multiple spindles for gang drilling. The most commonly used drill press is a single-spindle, floor-mounted, belt-driven machine for nonproduction drilling.

To switch out drill bits in a drill press, another tool called the chuck key is required. Insert the chuck key into one of the three holes in the chuck so that its teeth fit inside the grooves of the chuck. Turn the key to the left to loosen the chuck.

Figure 5: Chuck key for loosening and tightening a drill press chuck.

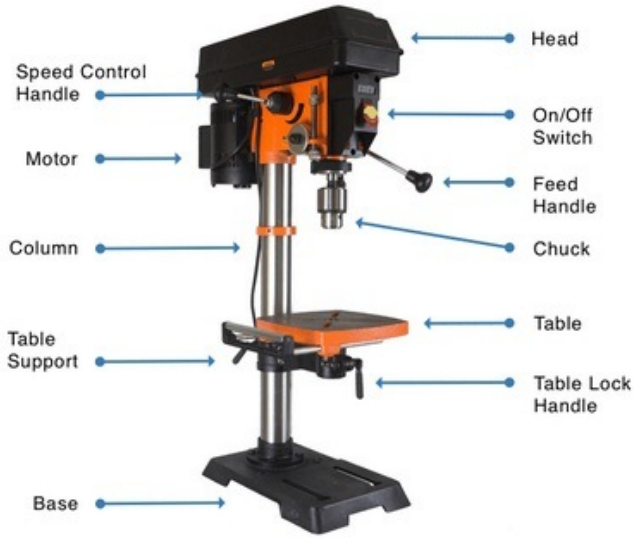

Figure 6: Main components of a drill press.

To operate a drill press, follow these steps from *toolnerds.com*:

Step #1: **Use Your PPE.** Protect yourself from projections such as splinters and chips with personal protective equipment. These include putting on a mask, wearing protective goggles, and donning a pair of rubber hand gloves, if needed.

Step #2: **Insert the Bit.** Insert the selected bit straight into the chuck and tighten. Ensure the key is removed, if the drill has a chuck key, before turning on the drill. To replace a bit, loosen the chuck by rotating it counterclockwise. Do not attempt this when the machine is operational.

Step #3: **Securely Clamp the Workpiece and/or the Drill.** Clamp the workpiece on the drill press table using a vice.

Step #4: **Adjust the Table Height.** Adjust the table and orient it, if movable, to the desired height that will be comfortable. When done, lock it in place.

Step #5: **Set Your Speed.** Adjust the speed of rotation according to the material and the diameter of the hole being drilled. Typically, drill press set-up speed is altered by moving the drive belt.

Step #6: Plug the Device. After checking the device's condition, plug in the power cord, then turn on the tool.

Step #7: Drill. With the lever, gently lower the bit toward the target on the workpiece. Let the drill do the work. Stop the engine once the hole is drilled and unplug the device from the mains.

3.4 Tap and Die

Instead of drilling clean holes, tap and die is a tool set that creates threaded holes and threaded rods. A tap is a threaded tool that cuts or forms threads in the female portion of a mating pair. Therefore, a tap creates threads inside a hole. A die is a threaded tool that cuts or forms threads in the male portion of a mating pair. Therefore, a die creates threads on a cylindrical rod surface to create a screw or bolt. Both taps and dies create new threads or repair damaged threads.

Figure 7: Tap and die.

To operate or use a tap and die, follow these steps from *tameson.com*:

Prepare the hole

1. This is the first step of tapping. It is the formation of a hole, usually by drilling, to a diameter smaller than the tap's outer diameter. This diameter is called the tap drill size.
2. The drill bit to be used for the hole diameter is listed in the drill and tap chart, usually inside the tap and die set. It is also available in machine shops.
3. Drill the hole, either a blind hole or through-hole.
4. Ensure removal of all chips from the hole.
5. Select tap and ratchet. Use the chart again to select the right tap size for the project. Remember, there are different types of taps. You may have to start with a tapered tap before finishing with a plug tap.
6. Depending on the tap size, use the appropriate ratchet. There are typically two sizes. Using the larger ratchet for a small tap may lead to excessive force that can break the tap.

Create the threads by hand tap

1. Fasten down the material to be threaded with a vice or clamp.
2. Insert the tap, ensure it is centered and aligned, and turn in the clockwise direction, making two complete rotations. Then turn anticlockwise for half a cycle. Repeat this

process until complete threading. This style of tapping prevents broken material from damaging the new threads.

3. Apply cutting fluid throughout the process to reduce heat and transport chips.
4. Keep the material and tap centered and level.
5. If restrictive force is felt, turn anticlockwise for half a rotation and try again. Do not force the thread, or the tap may break or incorrect threads may be made.
6. Remove the tap. Turn the ratchet anticlockwise until the tap is entirely out. The tapping process is complete.

How to use a die

The process of threading with a die involves several steps:

1. Prepare the rod. The material diameter should be slightly smaller than the die's major diameter. The die has a chamfered side used as the entry side for the rod. The chamfer helps center the die on the rod and reduces the start turning force when threading.
2. Select die and wrench. Like the tapping process, use the chart to select the right die with an appropriately sized wrench.
3. Create the threads. Fasten down the material with a vice or clamp.
4. Place the die on the material with the chamfered side starting the threads. Center and align the die, then turn in the clockwise direction, making two complete rotations.
5. Then turn anticlockwise for half a cycle. Repeat this process until the threading is complete. Use cutting fluid as required.
6. Remove the die. Turn the wrench anticlockwise until the die is entirely out. The threading process is complete.

4 References

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